

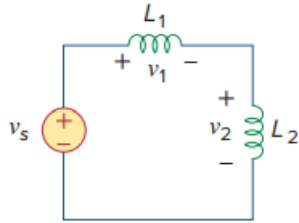
Problem

(a) For two inductors in series as in Fig. 6.81(a), show that the voltage division principle is

$$v_1 = \frac{L_1}{L_1 + L_2} v_s, \quad v_2 = \frac{L_2}{L_1 + L_2} v_s$$

assuming that the initial conditions are zero.

Fig. 6.81(a):

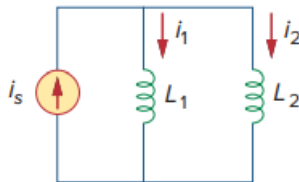


(b) For two inductors in parallel as in Fig. 6.81(b), show that the current-division principle is

$$i_1 = \frac{L_2}{L_1 + L_2} i_s, \quad i_2 = \frac{L_1}{L_1 + L_2} i_s$$

assuming that the initial conditions are zero.

Fig. 6.81(b):



Step-by-step solution

Step 1 of 2

$$(A) \quad V_s = (L_1 + L_2) \frac{di}{dt}$$

$$\frac{di}{dt} = \frac{V_s}{L_1 + L_2}$$

$$V_1 = L_1 \frac{di}{dt}, \quad V_2 = L_2 \frac{di}{dt}$$

$$V_1 = \frac{L_1}{L_1 + L_2} V_s, \quad V_L = \frac{L_2}{L_1 + L_2} V_s$$

Step 2 of 2

$$(B) \quad V_i = V_2 = L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt}$$

$$i_s = i_1 + i_2$$

$$\frac{di_s}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} = \frac{V}{L_1} + \frac{V}{L_2} = V \frac{(L_1 + L_2)}{L_1 L_2}$$

$$i_1 = \frac{1}{L_1} \int V dt = \frac{1}{L_1} \int \frac{L_1 L_2}{L_1 + L_2} \frac{di_s}{dt} dt = \frac{L_2}{L_1 + L_2} i_s$$

$$i_2 = \frac{1}{L_2} \int V dt = \frac{1}{L_2} \int \frac{L_1 L_2}{L_1 + L_2} \frac{di_s}{dt} dt = \frac{L_1}{L_1 + L_2} i_s.$$